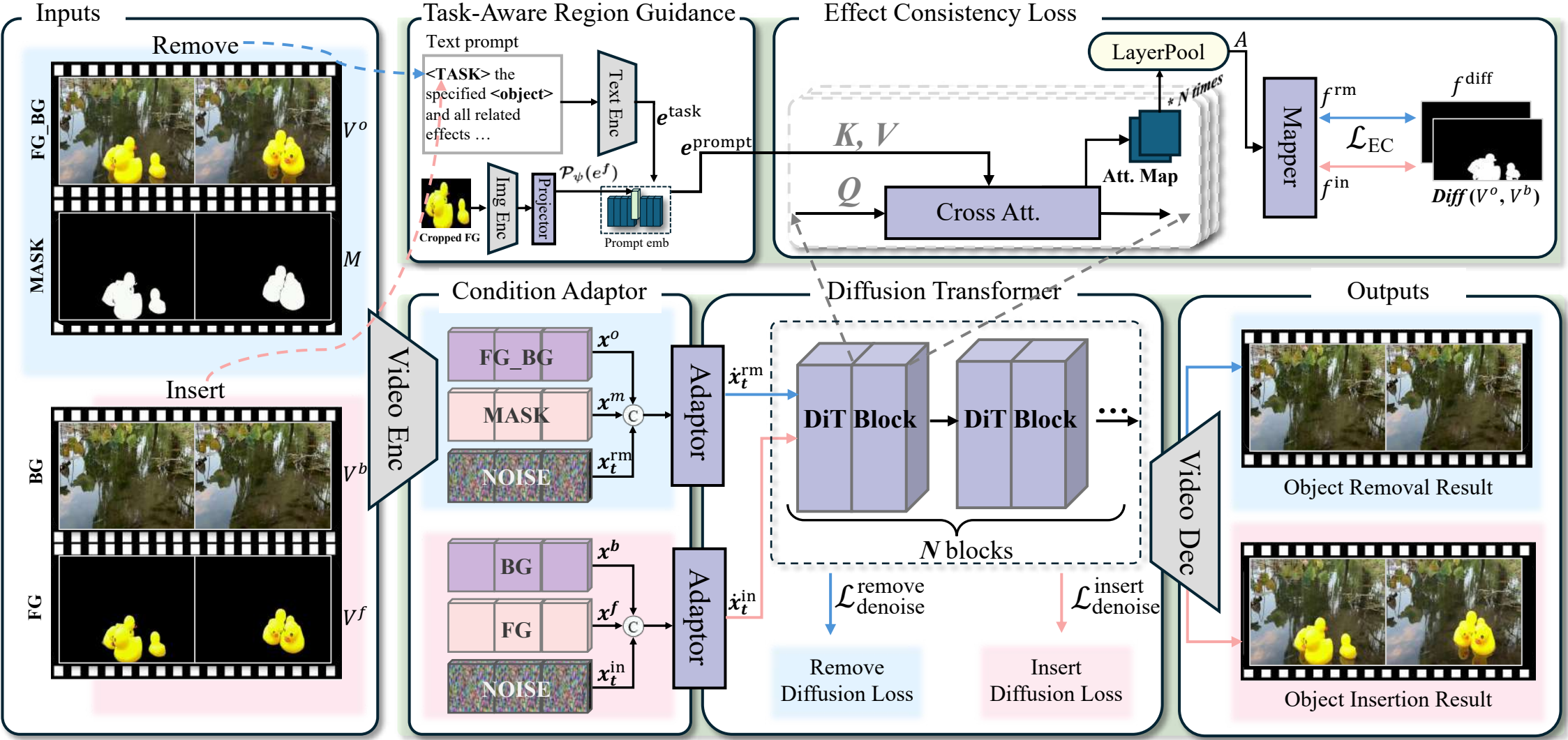
Inputs
Remove
FG_BG
MASK
V^o
M
Insert
BG
FG
V^b
V^f
Video Enc
Task-Aware Region Guidance
Text prompt
<TASK> the specified <object> and all related effects …
Text Enc
e^{task}
e^{prompt}
Cropped FG
Img Enc
Projector
$\mathcal{P}_\psi(e^f)$
Prompt emb
Effect Consistency Loss
LayerPool
A
$*N$ times
K, V
Q
Cross Att.
Att. Map
Mapper
f^{rm}
f^{in}
$\mathcal{L}_{\text{EC}}$
f^{diff}
$\textit{Diff}(V^o, V^b)$
Condition Adaptor
FG_BG
MASK
NOISE
x^o
x^m
x_t^{rm}
BG
FG
NOISE
x^b
x^f
x_t^{in}
Adaptor
Adaptor
$\dot{x}_t^{\text{rm}}$
$\dot{x}_t^{\text{in}}$
Diffusion Transformer
DiT Block
DiT Block
N blocks
$\mathcal{L}_{\text{denoise}}^{\text{remove}}$
$\mathcal{L}_{\text{denoise}}^{\text{insert}}$
Remove Diffusion Loss
Insert Diffusion Loss
Video Dec
Outputs
Object Removal Result
Object Insertion Result